

Exp 2 DDL Commands:

```
mysql> create Database Ott;
```

```
mysql> use Ott;
```

```
mysql> create table Student(roll int,name varchar(20),age int);
```

```
mysql> insert into Student values(1,"Salman",20);
```

```
mysql> select * from Student;
```

```
1 row in set (0.01 sec)
```

```
mysql> alter table Student add deptname varchar(20);
```

```
mysql> describe Student;
```

```
mysql> select * from Student;
```

```
mysql> alter table Student drop deptname;
```

```
mysql> select * from Student;
```

```
mysql> rename table Student to Stud;
```

```
mysql> select * from Stud;
```

```
mysql> truncate table Stud;
```

```
mysql> describe Stud;
```

```
mysql> select * from Stud;
```

```
mysql> drop table Stud;
```

```
mysql> drop database Ott;
```

Exp 3 DML Commands:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(roll int,name varchar(20),dept varchar(20),sal int);
```

```
insert into Stud values(7,"MSD","CSE",100);
```

```
insert into Stud values(18,"Virat","Civil",10);
```

```
insert into Stud values(21,"Hardik","AIDS",50);
```

```
select name,dept from Stud;
```

```
select name,sal from Stud where roll=21;
```

```
select * from Stud;
```

```
update Stud set name="Chiku" where roll=18;
```

```
select * from Stud;
```

```
delete from Stud where sal=10;
```

Exp 4 Pattern Matching & Aggregate functions:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(roll int,name varchar(20),dept varchar(20),sal int);
```

```
insert into Stud values(7,"Mahi","CSE",100);
```

```
insert into Stud values(18,"Virat","Civil",10);
```

```
insert into Stud values(21,"Hardik","AIDS",50);
```

```
insert into Stud values(16,"Max","IT",1000);
```

```
select * from Stud;
```

```
// Aggregate Functions:
```

```
select sum(sal) from Stud;
```

```
select max(sal) from Stud;
```

```
select min(sal) from Stud;
```

```
select avg(sal) from Stud;
```

```
select count(*) from Stud;
```

```
// Pattern Matching :
```

select name from Stud where name like "M%"; //(staring letter M)

select name from Stud where name like "%k"; //(lasting letter k)

select name from Stud where name like "____"; //(no of characters)

select * from Stud where name like "_a%"; //(second letter a)

select name from Stud where name like "%a%"; //(a letter between)

select name from Stud where name like "M%x"; //(staring with M and ending with x)

Exp 5 Join Operations & union,intersect,difference :

create Database Ott;

use Ott;

create table A(id int,name varchar(20));

insert into A values(1,"Rahul");

insert into A values(2,"Amit");

insert into A values(3,"Sneha");

insert into A values(4,"Neha");

select * from A;

```
create table B(id int,course varchar(20));
```

```
insert into B values(1,"AIDS");
```

```
insert into B values(2,"CSE");
```

```
insert into B values(3,"AIML");
```

```
insert into B values(5,"CYBER");
```

```
select A.id,A.name,B.course from A inner join B on A.id = B.id;
```

```
select A.id, A.name, B.course from A left join B on A.id = B.id;
```

```
select A.id, A.name, B.course from A right join B on A.id = B.id;
```

```
select * from A natural join B;
```

```
select A.id, A.name, B.course from A left join B on A.id = B.id
```

```
union
```

```
select A.id, A.name, B.course from A right join B on A.id = B.id;
```

```
// Relational Algebra operators.
```

```
select id from A union select id from B;
```

```
select id from A intersect select id from B;
```

```
select A.id from A inner join B on A.id = B.id;
```

Exp 6 Nested Queries in SQL:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(id int,name varchar(20),mark int,div varchar(20));
```

```
insert into Stud values(1,"Rahul",85,"A");
```

```
insert into Stud values(2,"Amit",95,"A");
```

```
insert into Stud values(3,"Sneha",89,"B");
```

```
insert into Stud values(4,"Neha",99,"B");
```

```
insert into Stud values(5,"Virat",60,"A");
```

```
insert into Stud values(6,"Pawan",70,"B");
```

```
select * from Stud;
```

```
// Single row Subquery:
```

```
select name from Stud where mark = (select max(mark) from Stud);
```

```
// Multi row Subquery:
```

```
select name from Stud where mark in(select mark from Stud where mark>80);
```

```
// Correlated Subquery:
```

```
select name from Stud S1 where mark > (select min(mark) from Stud S2 where S1.div =  
S2.div);
```

Exp 7 Create Views in SQL:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(id int,name varchar(20),mark int,div varchar(20));
```

```
insert into Stud values(1,"Rahul",85,"A");
```

```
insert into Stud values(2,"Amit",95,"A");
```

```
insert into Stud values(3,"Sneha",89,"B");
```

```
insert into Stud values(4,"Neha",99,"B");
```

```
insert into Stud values(5,"Virat",60,"A");
```



```
insert into Stud values(6,"Pawan",70,"B");
```

```
select * from Stud;
```

```
create view Studv as select name,id from Stud where div="A";
```

```
select * from Studv;
```

```
update Studv set name="Rakesh" where id=1;
```

```
delete from Studv where name="Virat";
```

```
drop view Studv;
```

Exp 8 Procedure & Trigger:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(id int,name varchar(20),mark int);
```

```
insert into Stud values(1,"Rahul",85);
```

```
insert into Stud values(2,"Amit",95);
```

```
insert into Stud values(3,"Neha",99);
```

```
select * from Stud;
```

```
mysql> delimiter $$
```

```
mysql> create procedure Highmarks()
```

```
-> begin
```

```
-> select * from Stud where mark>80;
```

```
-> end $$
```

Exp 9 DCL & TCL Commands:

```
create Database Ott;
```

```
use Ott;
```

```
create table Stud(id int,name varchar(20),mark int);
```

```
insert into Stud values(1,"Rahul",85);
```

```
insert into Stud values(2,"Amit",95);
```

```
insert into Stud values(3,"Neha",99);
```

```
select * from Stud;
```

```
// DCL Commands:
```

```
select user();
```

```
create user peter@localhost identified by "7";
```

```
show grants for peter@localhost;
```

```
grant select,update on Stud to peter@localhost;
```

```
show grants for peter@localhost;
```

```
revoke update on Stud from peter@localhost;
```

```
show grants for peter@localhost;
```

```
// TCL Commands:
```

```
(Rollback example):
```

```
start transcation;
```

```
update Stud set mark=50 where id=1;
```

```
select * from Stud;
```

```
rollback;
```

```
select * from Stud;
```

(Commit Example):

```
start transcation;
```

```
update Stud set mark=60 where id=1;
```

```
commit;
```

```
select * from Stud;
```

(Savepoint Example):

start transcation;

savepoint a;

update Stud set mark = 30 where id = 2;

ROLLBACK TO A;

select * from Stud;